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Individual Challenge Report - Data Cleaning Expert

Summary:

This report consists of an extensive data cleaning initiative focused on simulated ***GlobalWeatherRepository.csv*** which is one of the datasets of our group project. For this individual challenge, I created a messy data and intentionally injected every kind of complex data issues. Ranging from inconsistencies to obvious mistakes to mirror the real-world challenges. A carefully designed, multi-step process was created and used to clean the data, applying advanced methods to ensure accuracy and consistency.

The Challenge: Creating Real-World Data Messiness

The foundation of this challenge involved intentionally corrupting a good dataset to create a robust data cleaning pipeline. The challenge started by taking a clean dataset and intentionally making it messy. This was done to test how well-advanced data cleaning techniques could fix real-world data problems. The messiness was based on Data Quality Framework, which looks at five areas: Accuracy, Timeliness, Completeness, Relevance and Trustworthiness.

To simulate real-world data issues, a clean dataset was intentionally corrupted in multiple ways. Hidden missing values like "-" and "N/A" were inserted into key fields, and numeric columns were polluted with text (e.g., "82F", "Freezing"). Date formats were inconsistent, coordinates were altered or swapped, and outliers such as 1000°C or logically incorrect values like PM2.5 > PM10 were introduced. Duplicate records included both exact and near matches (e.g., "NYC" vs. "New York City"), and spelling mistakes, random capitalizations, and fake entries like "Test City" were added. Dates were also tampered with, showing years far in the future or past, making time analysis unreliable. These changes created a realistic testbed for evaluating the effectiveness of data cleaning techniques.

The Solution: A Step-by-Step Cleaning Process

To clean the dataset, a structured pipeline was followed. Hidden missing values were identified and standardized as proper nulls. Data types and formats were corrected using regular expressions, including cleaning units from numbers and fixing date strings. Missing values were imputed using the median, while categorical missing entries were removed. Outliers were detected using the IQR method and domain knowledge and corrected or removed. Duplicate entries were handled through exact match removal and fuzzy matching for near duplicates. Text fields were normalized for spelling, capitalization, and spacing. Dates were validated and corrected, keeping the most recent version where conflicts existed. Finally, irrelevant entries like "Test City" were removed to ensure a clean, consistent, and reliable dataset.

The Result: Cleaner, more Reliable Data

After cleaning, the dataset became much more reliable and usable. Below is the table summarizing *few* of the data cleaning steps.

Step	Description	Details / Counts
Initial Missing Data	Count of missing (NaN) values before cleaning	15,181
Imputation (Median Values)	Replaced missing values with median	Replaced values as follows: temperature- 22.0 precip- 0.0 humidity- 75.0 PM2.5- 7.5 PM10- 12.6 latitude- 17.25 longitude- 23.24
Missing Categorical Handling	Dropped rows with missing or "Unknown" country or location	787 ('Country'), 781 ('location_name')
Timestamp Cleaning	Dropped rows with invalid 'last_updated' timestamps	New shape: (38,249 × 10)
Outlier Detection & Capping	IQR method used to cap outliers to upper/lower bounds	temperature: 1,249; precip: 8,744; humidity: 1,159; PM2.5: 4,352; PM10: 5,173
Relational Outlier Correction	PM2.5 values exceeding PM10 were corrected	1,627 corrections
Duplicate Handling	Removed exact and near-duplicate records	63 exact; ~37,942 near-duplicates
Temporal Data Cleaning	Removed records outside valid date range (1998–2024) and kept most recent per location/day	5 removed; final shape: (228 × 11)